

Curcumin induces apoptosis in human non-small cell lung cancer NCI-H460 cells through ER stress and caspase cascade- and mitochondria-dependent pat...

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It has been reported that curcumin inhibited various types of cancer cells in vitro and in vivo. However, mechanisms of curcumin-inhibited cell growth and -induced apoptosis in human non-small cell lung cancer cells (NCI-H460) still remain unclear. In this study, NCI-H460 cells were treated with curcumin to determine its anticancer activity. Different concentrations of curcumin were used for different durations in NCI-H460 cells and the subsequent changes in the cell morphology, viability, cell cycle, mRNA and protein expressions were determined. Curcumin induced apoptotic morphologic changes in NCI-H460 cells in a dose-dependent manner. After curcumin treatment, BAX and BAD were up-regulated, BCL-2, BCL-X(L) and XIAP were down-regulated. In addition, reactive oxygen species (ROS), intracellular Ca^{2+} and endoplasmic reticulum (ER) stress were increased in NCI-H460 cells after exposure to curcumin. These signals led to a loss of mitochondrial membrane potential ($\Delta\text{Psi(m)}$) and culminated in caspase-3 activation. Curcumin-induced apoptosis was also stimulated through the FAS/caspase-8 (extrinsic) pathway and ER stress proteins, growth arrest- and DNA damage-inducible gene 153 (GADD153) and glucose-regulated protein 78 (GRP78) were activated in the NCI-H460 cells. Apoptotic cell death induced by curcumin was significantly reversed by pretreatment with ROS scavenger or caspase-8 inhibitor. Furthermore, the NCI-H460 cells tended to be arrested at the G(2)/M cell cycle stage after curcumin treatment and down-regulation of cyclin-dependent kinase 1 (CDK1) may be involved. In summary, curcumin exerts its anticancer effects on lung cancer NCI-H460 cells through apoptosis or cell cycle arrest.